

Displays

D1:

$$|\tau_1(t) - \tau_2(t)| \leq \beta(\tau_1.fstate, \tau_2.fstate, t), \quad (1)$$

D2:

$$\forall(x, y) \in \Gamma. x \leq ay + b. \quad (2)$$

D3:

$$\beta(x_1, x_2, t) = |x_1 - x_2|Ke^{\gamma t}.$$

D4:

$$\forall t \in [0, T]. |\tau_1(t) - \tau_2(t)| \leq |\tau_1.fstate - \tau_2.fstate|Ke^{\gamma t}.$$

D5:

$$\forall t. \ln \frac{|\tau_1(t) - \tau_2(t)|}{|\tau_1.fstate - \tau_2.fstate|} \leq \gamma t + \ln K.$$

D6:

$$\beta(x_1, x_2, t) = |x_1 - x_2|Ke^{\sum_{j=1}^{i-1} \gamma_j(t_j - t_{j-1}) + \gamma_i(t - t_{i-1})} \quad \text{for } t \in [t_{i-1}, t_i].$$

Inline

I1: $\mathbf{L}$

I2: n

I3: G

I4: $\mathcal{TL}$

I5: $\mathbb{R}^n$

I6: $\ell \in \mathbf{L}$

I7: ℓ

I8: $\mathbb{R}^2$

I9: G_1

I10: G_2

I11: `Powertrn`

I12: p

I13: λ

I14: pe

I15: i

I16: `startup`

I17: `normal`

I18: `powerup`

I19: `sensorfail`

I20: $G = \langle \mathbf{L}, \mathcal{V}, \mathcal{E}, vlab, elab \rangle$

I21: $\mathcal{V}$

I22: $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$

I23: $vlab : \mathcal{V} \rightarrow \mathbf{L}$

I24: $elab : \mathcal{E} \rightarrow \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0}$

I25: $\mathcal{V}_{\text{init}} \subseteq \mathcal{V}$

I26: $\mathcal{V}_{\text{term}} \subseteq \mathcal{V}$

I27: $\mathbf{L}_{\text{init}} = \{\ell \mid \exists v \in \mathcal{V}_{\text{init}}, vlab(v) = \ell\}$

I28: $\pi = v_1, t_1, v_2, t_2, \dots, v_k$

I29: $v_1 \in \mathcal{V}_{\text{init}}$

I30: $v_k \in \mathcal{V}_{\text{term}}$

I31: (v_i, t_i, v_{i+1})

I32: $(v_i, v_{i+1}) \in \mathcal{E}$

I33: $t_i \in elab((v_i, v_{i+1}))$

I34: Paths_G

I35: $vlab(\pi)$

I36: $vlab(v_1), t_1, vlab(v_2), t_2, \dots, vlab(v_k)$

I37: Trace_G
 I38: Powertrain
 I39: $\mathcal{V} = \{0, \dots, 4\}$
 I40: vlab
 I41: elab
 I42: $G_1, G_2,$
 I43: $\mathbf{L}_1, \mathbf{L}_2,$
 I44: $\text{lmap} : \mathbf{L}_1 \rightarrow \mathbf{L}_2$
 I45: $\sigma = \ell_1, t_1, \ell_2, t_2, \dots, \ell_k \in \text{Trace}_{G_1}$
 I46: $\text{lmap}(\sigma)$
 I47: $\text{lmap}(\ell_1), t_1, \text{lmap}(\ell_2), t_2, \dots, \text{lmap}(\ell_k)$
 I48: $G_1 \preceq_{\text{lmap}} G_2$
 I49: $\sigma \in \text{Trace}_{G_1}$
 I50: $\sigma' \in \text{Trace}_{G_2}$
 I51: σ'
 I52: G_1, G_2
 I53: $R \subseteq \mathcal{V}_1 \times \mathcal{V}_2$
 I54: $v \in \mathcal{V}_{\text{init}}$
 I55: $u \in \mathcal{V}_{\text{2init}}$
 I56: $(v, u) \in R$
 I57: $\text{lmap}(\text{vlab}_1(v)) = \text{vlab}_2(u)$
 I58: $(v, v') \in \mathcal{E}_1$
 I59: $(v, u) \in R$
 I60: $u_1, \dots, u_k$
 I61: u_j
 I62: $(v, u_j) \in R$
 I63: $\text{elab}_1((v, v')) \subseteq \cup_j \text{elab}_2((u, u_j))$
 I64: lmap
 I65: $|\mathcal{V}_{\text{init}}| = |\mathcal{V}_{\text{term}}| = 1$
 I66: $\mathcal{V}_{\text{init}}$
 I67: v_{init}
 I68: $\mathcal{V}_{\text{term}}$
 I69: v_{term}
 I70: $G_1 = \langle \mathbf{L}, \mathcal{V}_1, \mathcal{E}_1, \text{vlab}_1, \text{elab}_1 \rangle$
 I71: $G_2 = \langle \mathbf{L}, \mathcal{V}_2, \mathcal{E}_2, \text{vlab}_2, \text{elab}_2 \rangle$
 I72: $\text{vlab}_1(v_{\text{1term}}) = \text{vlab}_2(v_{\text{2init}})$
 I73: $G_1 \circ G_2 = \langle \mathbf{L}, \mathcal{V}, \mathcal{E}, \text{vlab}, \text{elab} \rangle$
 I74: $\mathcal{V} = (\mathcal{V}_1 \cup \mathcal{V}_2) \setminus \{v_{\text{2init}}\}$
 I75: $\mathcal{E} = \mathcal{E}_1 \cup \{(v_{\text{1term}}, u) \mid (v_{\text{2init}}, u) \in \mathcal{E}_2\} \cup \{(v, u) \in \mathcal{E}_2 \mid v \neq v_{\text{2init}}\}$
 I76: $\text{vlab}(v) = \text{vlab}_1(v)$
 I77: $v \in \mathcal{V}_1$
 I78: $\text{vlab}(v) = \text{vlab}_2(v)$
 I79: $v \in \mathcal{V}_2$
 I80: $(v, u) \in \mathcal{E}$
 I81: $\text{elab}((v, u))$
 I82: $\text{elab}_1((v, u))$
 I83: $u \in \mathcal{V}_1$
 I84: $\text{elab}_2((v_{\text{2init}}, u))$
 I85: $v = v_{\text{1term}}$
 I86: $\text{elab}_2((v, u)), \text{otherwise}$
 I87: $G_1 \preceq_{\text{id}} G_1 \circ G_2$
 I88: id
 I89: $G_1 \circ G_2$
 I90: $\tau : [0, T] \rightarrow \mathbb{R}^n$
 I91: $T \geq 0$
 I92: $[0, T]$

I93: τ
 I94: $\tau.dom$
 I95: $\tau(0)$
 I96: $\tau.fstate$
 I97: $\tau.lstate = \tau(T)$
 I98: $\tau.ltime = T$
 I99: $\langle \tau, \ell \rangle$
 I100: T_1
 I101: $\langle \tau, \ell \rangle$
 I102: $T_1 \in \tau.dom$
 I103: $\langle \tau_1, \ell \rangle$
 I104: $\tau_1 : [0, T_1] \rightarrow \mathbb{R}^n$
 I105: $t \in [0, T_1]$
 I106: $\tau_1(t) = \tau(t)$
 I107: $\langle \tau, \ell \rangle \in \mathcal{TL}$
 I108: $\langle \tau_1, \ell_1 \rangle, \langle \tau_2, \ell_2 \rangle \in \mathcal{TL}$
 I109: $\tau_1.fstate = \tau_2.fstate$
 I110: $\ell_1 = \ell_2$
 I111: $\mathcal{TL}_{init, \ell} = \{\tau.fstate \mid \langle \tau, \ell \rangle \in \mathcal{TL}\}$
 I112: $\mathcal{TL}_{init, \ell}$
 I113: $\ell \in \mathbf{L}, T > 0$
 I114: $x \in \mathcal{TL}_{init, \ell}$
 I115: $\langle \tau, \ell \rangle \in \mathcal{TL}$
 I116: $\tau.fstate = x$
 I117: $\tau(\cdot)$
 I118: sim
 I119: $x_0 \in \mathcal{TL}_{init, \ell}$
 I120: $t_1, \dots, t_k$
 I121: $sim(x_0, \ell, t_1), \dots, sim(x_0, \ell, t_k)$
 I122: $\langle \tau, \ell \rangle \in \mathcal{TL}$
 I123: $\tau.fstate = x_0$
 I124: $i \in \{1, \dots, k\}$
 I125: $sim(x_0, \ell, t_i) = \tau(t_i)$
 I126: $\mathcal{TL}_1$
 I127: $\mathbf{L}_1$
 I128: $\mathcal{TL}_2$
 I129: $\mathbf{L}_2$
 I130: $\langle \tau, \ell \rangle \in \mathcal{TL}_1$
 I131: $lmap(\langle \tau, \ell \rangle)$
 I132: $\langle \tau, lmap(\ell) \rangle$
 I133: $\mathcal{TL}_1 \preceq_{lmap} \mathcal{TL}_2$
 I134: $lmap(\langle \tau, \ell \rangle) \in \mathcal{TL}_2$
 I135: $\mathcal{H}$
 I136: $\langle \mathbf{L}, \Theta, G, \mathcal{TL} \rangle$
 I137: $\Theta \subseteq \mathbb{R}^n$
 I138: $G = \langle \mathbf{L}, \mathcal{V}, \mathcal{E}, elab \rangle$
 I139: $\mathbb{R}^n \times \mathbf{L}$
 I140: $\Theta \times \mathbf{L}_{init}$
 I141: $\alpha = \langle \tau_1, \ell_1 \rangle \dots, \langle \tau_{k-1}, \ell_{k-1} \rangle, \ell_k$
 I142: $\tau_1.fstate \in \Theta$
 I143: $\ell_1 \in \mathbf{L}_{init}$
 I144: $path(\alpha)$
 I145: $\ell_1, \tau_1.ltime, \ell_2, \dots, \ell_k$
 I146: $\tau_{i+1}.fstate = \tau_i.lstate$
 I147: $Execs_{\mathcal{H}}$
 I148: $\alpha.fstate = \tau_1.fstate$

I149: $\alpha.lstate = \tau_{k-1}.lstate$
 I150: $\alpha.fmode = \ell_1$
 I151: $\alpha.lmode = \ell_k$
 I152: $\langle x, \ell \rangle$
 I153: t
 I154: v
 I155: $\alpha = \langle \tau_1, \ell_1 \rangle \dots, \langle \tau_{k-1}, \ell_{k-1} \rangle, \ell_k \in \text{Execs}_{\mathcal{H}}$
 I156: $\pi = v_1, t_1, \dots, v_k$
 I157: $i \in \{1, \dots, k\}$
 I158: $t' \in \tau_i.dom$
 I159: $vlab(\pi) = \text{path}(\alpha)$
 I160: $v = v_i$
 I161: $\ell = \ell_i$
 I162: $x = \tau_i(t')$
 I163: $t = t' + \sum_{j=1}^{i-1} t_j$
 I164: $\text{ReachTube}_{\mathcal{H}} = \{ \langle x, \ell, t \rangle \mid \text{for some } v, \langle x, \ell \rangle \text{ is reachable at time } t \text{ and vertex } v \}$
 I165: $\text{Reach}_{\mathcal{H}} = \{ \langle x, \ell \rangle \mid \text{for some } v, t, \langle x, \ell \rangle \text{ is reachable at time } t \text{ and vertex } v \}$
 I166: $\text{Reach}_{\mathcal{H}}^v = \{ \langle x, \ell \rangle \mid \text{for some } t, \langle x, \ell \rangle \text{ is reachable at time } t \text{ and vertex } v \}$
 I167: $\mathcal{U} \subseteq \mathbb{R}^n \times \mathbb{L}$
 I168: $\text{Reach}_{\mathcal{H}} \cap \mathcal{U} = \emptyset$
 I169: $vlab(v)$
 I170: $\mathcal{U}_p$
 I171: $t > 4 \wedge \lambda \notin [12.4, 12.6]$
 I172: $t > 4 \wedge \lambda \notin [14.6, 14.8]$
 I173: $\mathcal{H}_i = \langle \mathbb{L}_i, \Theta_i, G_i, \mathcal{TL}_i \rangle$
 I174: $i \in \{1, 2\}$
 I175: $lmap : \mathbb{L}_1 \rightarrow \mathbb{L}_2$
 I176: $\Theta_1 \subseteq \Theta_2$
 I177: $\text{Reach}_{\mathcal{H}_1} \subseteq \text{Reach}_{\mathcal{H}_2}$
 I178: x, y
 I179: `cruise`
 I180: `speedup`
 I181: `brake`
 I182: `em.brake`
 I183: `ch_left`
 I184: `ch_right`
 I185: sx
 I186: vx
 I187: x
 I188: sy
 I189: vy
 I190: y
 I191: `Merge`
 I192: B
 I193: `AutoPassing`
 I194: `Merge3`
 I195: `AEB`
 I196: $\text{ReachTube}_{\mathcal{H}}$
 I197: $\beta : \mathbb{R}^n \times \mathbb{R}^n \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$
 I198: $\langle \tau_1, \ell \rangle, \langle \tau_2, \ell \rangle \in \mathcal{TL}$
 I199: $t \in \tau_1.dom \cap \tau_2.dom$
 I200: β
 I201: 0
 I202: $t \in \tau.dom$
 I203: $\tau_1, \dots, \tau_k, \dots$
 I204: $\tau_k.fstate \rightarrow \tau.fstate$

I205: $\beta(\tau_k.fstate, \tau.fstate, t) \rightarrow 0$
 I206: $\Gamma \subseteq \mathbb{R} \times \mathbb{R}$
 I207: $(a, b) \in \mathbb{R}^2$
 I208: Γ
 I209: (a_*, b_*)
 I210: (a, b)
 I211: 1
 I212: $\mathcal{D}$
 I213: $\text{err}_{\mathcal{D}}(a, b) = \mathcal{D}(\{(x, y) \in \Gamma \mid x > ay + b\})$
 I214: $\mathcal{D}(X)$
 I215: X
 I216: k
 I217: $(x_1, y_1), \dots, (x_k, y_k)$
 I218: $x_i \leq ay_i + b$
 I219: $\epsilon, \delta \in \mathbb{R}_+$
 I220: $k \geq \frac{1}{\epsilon} \ln \frac{1}{\delta}$
 I221: $\geq 1 - \delta$
 I222: $\text{err}_{\mathcal{D}}(a, b) < \epsilon$
 I223: $> \epsilon$
 I224: $B = \{(x, y) \mid x > ay + b\}$
 I225: $\mathcal{D}(B) > \epsilon$
 I226: $\leq (1 - \epsilon)^k$
 I227: $(1 - s) \leq e^{-s}$
 I228: s
 I229: $(1 - \epsilon)^k \leq e^{-\epsilon k} \leq e^{-\ln \frac{1}{\delta}} = \delta$
 I230: $\square$
 I231: K
 I232: γ
 I233: τ_1
 I234: τ_2
 I235: $(\ln \frac{|\tau_1(t) - \tau_2(t)|}{|\tau_1.fstate - \tau_2.fstate|}, t)$
 I236: τ_1, τ_2
 I237: $\mathcal{TL}_{\text{init}, \ell}$
 I238: $\gamma T + \ln K$
 I239: $0 = t_0, t_1, \dots, t_N = T$
 I240: $K, \gamma_1, \gamma_2, \dots, \gamma_N$
 I241: $i \in \{1, \dots, N\}$
 I242: $t \in [t_{i-1}, t_i]$
 I243: $|\tau_1(t) - \tau_2(t)| \leq |\tau_1(t_{i-1}) - \tau_2(t_{i-1})| K e^{\gamma_i t}$
 I244: γ_i
 I245: Γ_i
 I246: t_{i-1}
 I247: t_i
 I248: ≥ 2
 I249: $\gamma_i < 2$
 I250: S_{train}
 I251: S_{test}
 I252: $|S_{\text{train}}| > 10$
 I253: $|S_{\text{train}}| > 20$
 I254: GraphReach
 I255: $\mathcal{H} = \langle \mathbf{L}, \Theta, G, \mathcal{TL} \rangle$
 I256: RS
 I257: $\langle RT, v \rangle$
 I258: $v \in \mathcal{V}$
 I259: RT
 I260: VerInit

I261: $\langle S, v \rangle$
 I262: $S \subset \mathbb{R}^n$
 I263: $RS = \emptyset$
 I264: $VerInit = \{\langle \Theta, v_{init} \rangle\}$
 I265: $LearnDiscrepancy(S_{init}, d, \ell)$
 I266: S_{init}
 I267: d
 I268: $ReachComp(S_{init}, d, \beta)$
 I269: $Order$
 I270: ptr
 I271: $curv$
 I272: dt
 I273: $nextv$
 I274: $elab((curv, nextv))$
 I275: $GraphReach(\mathcal{H})$
 I276: $RS \leftarrow \emptyset; VerInit \leftarrow \{\langle \Theta, v_{init} \rangle\}; Order \leftarrow TopSort(G)$
 I277: $ptr = 0 : len(Order) - 1$
 I278: $curv \leftarrow Order[ptr]$
 I279: $\ell \leftarrow vlab(curv)$
 I280: $dt \leftarrow \max\{t' \in \mathbb{R}_{\geq 0} \mid \exists vs \in \mathcal{V}, (curv, vs) \in \mathcal{E}, (t, t') \leftarrow elab((curv, vs))\}$
 I281: $S_{init} \in \{S \mid \langle S, curv \rangle \in VerInit\}$
 I282: $\beta \leftarrow LearnDiscrepancy(S_{init}, dt, \ell)$
 I283: $RT \leftarrow ReachComp(S_{init}, dt, \beta)$
 I284: $RS \leftarrow RS \cup \langle RT, curv \rangle$
 I285: $nextv \in curv.succ$
 I286: $(t, t') \leftarrow elab((curv, nextv))$
 I287: $VerInit \leftarrow VerInit \cup \langle Restr(RT, (t, t')), nextv \rangle$
 I288: $VerifySafety$
 I289: $\mathcal{U}$
 I290: Θ
 I291: S
 I292: $[14.6, 14.8]$
 I293: $[12.4, 12.6]$
 I294: $LearnDiscrepancy$
 I295: $VerifySafety(\mathcal{H}, U)$
 I296: sx_A, sx_B
 I297: sy_A, sy_B
 I298: $i \in \{A, B\}$
 I299: $sx_i = vx_i = 0$
 I300: $vy_i = 1$
 I301: $sy_A \in [0, 2], sy_B \in [15, 17]$
 I302: sx_A
 I303: sx_B
 I304: sy_A
 I305: sy_B
 I306: $(|sx_A - sx_B| < 2 \ \& \ |sy_A - sy_B| < 2)$
 I307: $sy_B \in [30, 40]$
 I308: $\mathcal{U}_c$
 I309: $\mathcal{U}_t$
 I310: sy_C
 I311: ATS
 I312: $\mathcal{H}_1$
 I313: $\mathcal{H}_2$
 I314: T
 I315: $\{R_i\}_{i \in I}$
 I316: $\cup_{i \in I} R_i$

I317: $\mathcal{V}_1 \times \mathcal{V}_2$
 I318: (v, u)
 I319: u
 I320: $|\mathcal{V}_1|$
 I321: G_i, G'_i
 I322: $G'_1 \circ G'_2$
 I323: $G_i \preceq_{\text{id}} G'_i$
 I324: $G_1 \circ G_2 \preceq_{\text{id}} G'_1 \circ G'_2$
 I325: G_1, G_2
 I326: $v_{1\text{term}}$
 I327: $\mathcal{H} = \langle \mathbf{L}, \Theta, G_1 \circ G_2, \mathcal{TL} \rangle$
 I328: $\mathcal{H}_1 = \langle \mathbf{L}, \Theta, G_1, \mathcal{TL} \rangle$
 I329: $\mathcal{H}_2 = \langle \mathbf{L}, \text{Reach}_{\mathcal{H}_1}^{v_{1\text{term}}}, G_2, \mathcal{TL} \rangle$
 I330: $\text{Reach}_{\mathcal{H}} = \text{Reach}_{\mathcal{H}_1} \cup \text{Reach}_{\mathcal{H}_2}$
 I331: $G \circ G$
 I332: $\mathcal{H}'$
 I333: $\text{Reach}_{\mathcal{H}} \subseteq \text{Reach}_{\mathcal{H}'}$
 I334: $i \geq 1$
 I335: $\mathcal{H}_i = \langle \mathbf{L}, \Theta, G^i, \mathcal{TL} \rangle$
 I336: G^i
 I337: $\mathcal{H}_1 = \langle \mathbf{L}, \Theta, G, \mathcal{TL} \rangle$
 I338: $\text{Reach}_{\mathcal{H}_1}^{v_{\text{term}}} \subseteq \Theta$
 I339: $\text{Reach}_{\mathcal{H}_i} \subseteq \text{Reach}_{\mathcal{H}_1}$
 I340: $\Theta_1 = \text{Reach}_{\mathcal{H}_1}^{v_{\text{term}}}$
 I341: $\Theta_1 \subseteq \Theta$
 I342: $\mathcal{H}'_i = \langle \mathbf{L}, \Theta_1, G^i, \mathcal{TL} \rangle$
 I343: $\text{Reach}_{\mathcal{H}'_i} \subseteq \text{Reach}_{\mathcal{H}_i}$
 I344: $i = 1$
 I345: $\circ$
 I346: $\text{Reach}_{\mathcal{H}_{i+1}} = \text{Reach}_{\mathcal{H}_1} \cup \text{Reach}_{\mathcal{H}'_i} \subseteq \text{Reach}_{\mathcal{H}_1} \cup \text{Reach}_{\mathcal{H}_i} = \text{Reach}_{\mathcal{H}_1}$
 I347: $G_1 \circ G_2^i \circ G_3$
 I348: $G_1 \circ G_2 \circ G_3$
 I349: $[1, 2]$
 I350: $[2.5, 3.5]$
 I351: $[0.5, 4.5]$
 I352: $G_2 \preceq_{\text{id}} G_1$
 I353: G_A
 I354: $(v_0, \text{startup}) \xrightarrow{[5,10]} (v_1, \text{normal}) \xrightarrow{[10,15]} (v_2, \text{powerup})$
 I355: G_B
 I356: $(v_0, \text{powerup}) \xrightarrow{[5,10]} (v_1, \text{normal}) \xrightarrow{[10,15]} (v_2, \text{powerup})$
 I357: $G_1 = (v_0, \text{startup}) \xrightarrow{[5,10]} (v_1, \text{normal}) \xrightarrow{[10,15]} (v_2, \text{powerup}) \xrightarrow{[5,10]} (v_3, \text{normal}) \xrightarrow{[10,15]} (v_4, \text{powerup})$
 I358: $G_1 = G_A \circ G_B$
 I359: $\mathcal{H}_i = \langle \mathbf{L}, \Theta_i, G_i, \mathcal{TL} \rangle$
 I360: $\Theta_A = \Theta$
 I361: $\Theta_B = \text{Reach}_{\mathcal{H}_A}^{v_2}$
 I362: Θ_B
 I363: $\text{Reach}_{\mathcal{H}_B}^{v_2}$
 I364: $\text{Reach}_{\mathcal{H}_B}^{v_2} \subseteq \Theta_B$
 I365: $\mathcal{H}_i = \langle \mathbf{L}, \Theta, G_i, \mathcal{TL} \rangle$
 I366: $G_A \circ G_B$
 I367: $G_2 = G_A \circ G_B \circ G_B \circ G_B$
 I368: δ
 I369: ω
 I370: MergeBehind
 I371: MergeAhead

I372: MergeBetween
 I373: gear1
 I374: gear2
 I375: gear3
 I376: gear4
 I377: T_i
 I378: T_o
 I379: > 4000
 I380: $VerifySafety(\mathcal{H}, \mathcal{U})$
 I381: $\mathcal{I}.push(Partition(\Theta))$
 I382: $\mathcal{I} \neq \emptyset$
 I383: $S \leftarrow \mathcal{I}.pop()$
 I384: $RS \leftarrow GraphReach(\mathcal{H})$
 I385: $RS \cap \mathcal{U} = \emptyset$
 I386: $\exists(x, l, t) \in RT$
 I387: $\langle RT, v \rangle \in RS$
 I388: $(x, l, t) \subseteq \mathcal{U}$
 I389: $\langle RT, v \rangle$
 I390: $\mathcal{I}.push(Partition(S))$
 I391: $G \leftarrow RefineGraph(G)$
 I392: L_{init}
 I393: $\mathcal{L}$
 I394: $\mathcal{E}$
 I395: $\tau.lstate$
 I396: $\tau.ltime$
 I397: $Reach_{\mathcal{H}}$
 I398: $err_{\mathcal{D}}$
 I399: $\{1, 2\}$
 I400: $|\tau_1(t) - \tau_2(t)|$
 I401: $|\tau_1(t) - \tau_2(t)|$
 I402: $\gamma_i t$
 I403: $\gamma_i(t - t_{i-1})$
 I404: U
 I405: $\leftarrow$
 I406: $\mathcal{I}$
 I407: I
 I408: l